Data Engineering Accelerated Program Course Syllabus

A practical four-week program for building production-style data pipelines, cloud warehouses, transformations, orchestration, CI/CD automation, and BI dashboards.

Duration	Portfolio	Outcome
4 weeks	**5 projects**	**Capstone platform**

Extract — **Store** — **Model** — **Orchestrate** — **Dashboard**

Capstone pattern: API/Python -> raw data -> Snowflake/dbt -> Airflow/GitHub Actions -> Power BI

Program rhythm

Week 1	Week 2	Week 3	Week 4
Python, Pandas, APIs, PostgreSQL	RDBMS, SQL, modeling, warehouse design	Snowflake, dbt, ELT, transformation	Airflow, CI/CD, Power BI, capstone

What learners build

- Sales and weather data pipelines with clean exports and analytical reports.
- A dimensional sales data warehouse with validation queries and performance-aware design.
- Snowflake and dbt transformation layers with tests, documentation, and Git publishing.
- End-to-end capstone pipelines scheduled in Airflow and delivered through Power BI.

Week 1
weekly course syllabus

Foundations, Python, and DataFrames

Course modules, project outcomes, and technologies used

Weekly course syllabus

Modules and practice are sequenced so each week produces a portfolio artifact and supports the final capstone.

Introduction to Data Engineering

Core concepts, examples, and guided practice.

Python for Data Engineers

Core concepts, examples, and guided practice.

Pandas

Core concepts, examples, and guided practice.

Mini Project 1 - Sales Data Pipeline

Core concepts, examples, and guided practice.

Visual tool stack for this week

- Py Python
- PG PostgreSQL
- pd Pandas
- SA SQLAlchemy
- pg2 psycopg2
- Jn Jupyter Notebook
- Rq Requests
- {} JSON

Project work

Mini Project 1 - Sales Data Pipeline

Build a pipeline that extracts sales data from PostgreSQL, cleans and validates it, creates metrics, generates reports, and exports processed outputs.

- Extract sales data from PostgreSQL.

- Clean and validate the data.

- Create calculated metrics.

- Generate reports.

- Export processed data.

Technologies used: Python, PostgreSQL, Pandas, SQLAlchemy, psycopg2, Jupyter Notebook

Mini Project 2 - Weather Data Pipeline

Build a complete ETL pipeline that extracts weather data from a public API, transforms it in Pandas, and saves polished analytical outputs.

- Extract data from a public API.

- Load data into a Pandas DataFrame.

- Clean and validate the data.

- Perform transformations and feature engineering.

- Generate summary statistics.

- Filter and sort records.

- Save cleaned data into CSV and JSON.

- Produce a simple analytical report.

Technologies used: Python, Jupyter Notebook, Pandas, Requests, JSON, CSV and JSON Files

Week 2
weekly course syllabus

Relational Design, SQL, and Warehousing

Course modules, project outcomes, and technologies used

Weekly course syllabus

Modules and practice are sequenced so each week produces a portfolio artifact and supports the final capstone.

Database Design - RDBMS

- What is RDBMS?
- SQL databases
- SQL DDL statements
- SQL DML statements
- SQL DQL statements
- SQL constraints
- SQL primary keys
- SQL foreign keys
- SQL composite keys
- SQL alternative keys
- Data integrity

Advanced SQL Concepts

- What is JOIN?
- Types of SQL JOIN
- SQL aggregate functions
- LIKE operator
- IN operator
- BETWEEN operator
- UNION
- GROUP BY
- ORDER BY
- HAVING
- EXISTS
- ALL and ANY
- Window functions

Dimension Modeling

- What is dimensional modeling?
- Fact tables
- Dimension tables
- Dimension keys
- Slowly changing dimensions
- Flat schema
- Star schema
- Snowflake schema
- Galaxy schema
- CTEs

Cardinality and Normalization

- What is normalization?
- Unnormalized form
- First normal form
- Second normal form
- Third normal form
- Boyce-Codd normal form
- Fourth normal form
- Fifth normal form
- Sixth normal form
- Cardinality
- Types of cardinality
- Index
- View

Visual tool stack for this week

SQL SQL
PG PostgreSQL

Project work

Mini Project - Sales Data Warehouse

Design a warehouse that turns normalized sales data into reporting-ready dimensional models and analytics outputs.

Data Modeling

- OLTP vs OLAP
- Normalized models
- Dimensional modeling

Data Warehousing

- Fact tables
- Dimension tables
- Star schema design

ETL / ELT

- Extract data
- Transform data
- Load data

SQL Engineering

- Joins
- Aggregations
- Window functions
- CTEs
- Data validation queries

Analytics Engineering

- Revenue reporting
- Customer analytics
- Product analytics
- Store performance analysis

Best Practices

- Surrogate keys
- Data quality checks
- Performance tuning
- Scalable warehouse design

Technologies used: SQL, PostgreSQL

Week 3
weekly course syllabus

Cloud Data Warehousing and Transformation

Course modules, project outcomes, and technologies used

Weekly course syllabus

Modules and practice are sequenced so each week produces a portfolio artifact and supports the final capstone.

Cloud Data Warehousing

- What is a cloud data platform?
- Components of a cloud data platform
- The cloud data landscape
- What is a data warehouse?
- ETL
- ELT
- ETL vs ELT

Snowflake

- What is Snowflake?
- Why Snowflake?
- Snowflake architecture
- Snowflake editions
- Virtual warehouses
- Snowflake Marketplace
- RBAC
- Loading data
- COPY INTO
- Snowpipe
- Databases and schemas
- Tables
- Time Travel

dbt

- What is dbt?
- dbt architecture
- dbt in the modern data stack
- Installing dbt
- dbt project structure
- Layered modeling
- Choosing models
- Jinja in dbt
- Model selection
- Running dbt
- dbt testing
- Documentation
- dbt Core vs dbt Cloud

Visual tool stack for this week

SF Snowflake

dbt dbt

SQL SQL

Git Git and GitHub

CSV CSV and JSON Fi...

Project work

Sales Data Warehousing and Transformation

Load raw CSV files into Snowflake, organize schemas, build dbt models, test quality, document lineage, and publish the project to GitHub.

- Load raw CSV files into Snowflake.

- Organize data into schemas.

- Build staging models using dbt.

- Build intermediate models.

- Build mart models.

- Test data quality.

- Document the project.

- Track changes using Git.

- Publish the project to GitHub.

Technologies used: Snowflake, dbt, SQL, Git and GitHub, CSV and JSON Files

Week 4
weekly course syllabus

Orchestration, CI/CD, BI, and Capstone Delivery

Course modules, project outcomes, and technologies used

Weekly course syllabus

Extract — **Store** — **Model** — **Orchestrate** — **Dashboard**

Capstone pattern: API/Python -> raw data -> Snowflake/dbt -> Airflow/GitHub Actions -> Power BI

Modules and practice are sequenced so each week produces a portfolio artifact and supports the final capstone.

Apache Airflow

- What is Apache Airflow?
- Airflow architecture
- Core components
- What is a DAG?
- Task
- Task dependencies
- Executors
- Alerting and notifications
- Scheduling
- Cron expression
- Airflow folder structure
- Monitoring and logging
- Airflow vs other tools

CI/CD for Data Projects with GitHub Actions

- What is CI/CD?
- Continuous integration
- Continuous delivery
- CI/CD tools
- What is GitHub Actions?
- GitHub Actions components
- Python CI workflow
- CI pipeline for dbt
- GitHub Actions for Snowflake
- GitHub Actions for Apache Airflow
- GitHub Actions pipeline for Airflow
- Code quality checks
- Branching strategy
- Pull request workflow
- Data engineering CI/CD pipeline

Business Analysis - Power BI

- What is business intelligence?
- Power BI
- Power BI architecture
- Visualizations
- Common charts
- Uncommon charts
- Filters
- Slicers
- Getting data in Power BI
- Publish
- Power BI workspace

Visual tool stack for this week

AF Apache Airflow	GA GitHub Actions	PBI Power BI	API REST APIs
Py Python	{} JSON	SF Snowflake	dbt dbt

Project work

Capstone Project 1 - API to Dashboard Platform

Build an end-to-end modern data platform from REST APIs through Snowflake, dbt, Airflow, GitHub Actions, and Power BI.

- Extract data from REST APIs using Python.
- Store and manage raw datasets following data lake principles.
- Load and query data efficiently in Snowflake.
- Build modular, testable transformation pipelines with dbt.
- Design and schedule end-to-end workflows using Apache Airflow.
- Create optimized reporting tables for analytics.
- Develop professional Power BI dashboards with actionable insights.
- Implement CI/CD automation using GitHub Actions.
- Apply layered architectures using Raw/Bronze, Silver, and Gold data zones.
- Use version control, testing, and documentation as delivery standards.

Technologies used: REST APIs, Python, Snowflake, dbt, Apache Airflow, GitHub Actions, Power BI

Capstone Project 2 - Complete Analytics Warehouse

Consume APIs, work with JSON, design Snowflake databases, transform data with dbt, schedule Airflow pipelines, and publish BI dashboards.

- Consume REST APIs.
- Work with JSON data.
- Build ETL/ELT pipelines.
- Design Snowflake databases.
- Transform data using dbt.
- Build star schemas.
- Create fact and dimension tables.
- Schedule pipelines with Airflow.
- Build Power BI dashboards.
- Implement CI/CD using GitHub Actions.

Technologies used: REST APIs, JSON, Snowflake, dbt, Apache Airflow, Power BI, GitHub Actions

Tool Glossary

Visual guide to the program tool stack.

Each tool appears with a compact visual badge, where it fits in the course, and the practical job it performs in the data engineering workflow.

Py
Python
Primary language for extracting API data, cleaning records, validating pipelines, and automating data engineering tasks.

Weeks 1 and 4

PG
PostgreSQL
Relational database used to store source sales data, practice SQL, and understand transactional systems.

Weeks 1 and 2

pd
Pandas
Python library for DataFrames, cleaning, validation, transformations, feature engineering, and reports.

Week 1

SA
SQLAlchemy
Python toolkit that creates database connections and helps pipelines read from relational sources.

Week 1

pg2
psycopg2
PostgreSQL driver used by Python code to connect, query, and extract data from PostgreSQL.

Week 1

Jn
Jupyter Notebook
Interactive notebook environment for exploration, profiling, documentation, and small pipeline prototypes.

Week 1

Rq
Requests
Python HTTP library used to call public APIs, manage responses, and feed raw JSON into a pipeline.

Week 1

{}
JSON
Common API payload format used for nested records, raw files, transformations, and REST integrations.

Weeks 1 and 4

SQL
SQL
Core language for querying, joining, aggregating, validating, and designing analytical data models.

Weeks 2 and 3

SF
Snowflake
Cloud data warehouse for loading raw data, organizing schemas, querying at scale, and serving marts.

Weeks 3 and 4

dbt
dbt
Analytics engineering framework for modular SQL models, tests, documentation, lineage, and transformations.

Weeks 3 and 4

AF
Apache Airflow
Workflow orchestrator for DAGs, dependencies, schedules, retries, monitoring, and production pipelines.

Week 4

GA

GitHub Actions

CI/CD automation for Python checks, dbt validation, pull request workflows, and deployment gates.

Week 4

PBI

Power BI

Business intelligence platform for dashboards, charts, slicers, publishing, and decision-ready reporting.

Week 4

API

REST APIs

External data source pattern used to consume live data, work with HTTP responses, and build ingestion jobs.

Weeks 1 and 4

Git

Git and GitHub

Version control and collaboration workflow for dbt projects, code review, branching, and portfolio publishing.

Weeks 3 and 4

CSV

CSV and JSON Files

Portable file formats for extracts, raw layers, cleaned outputs, exports, and pipeline handoffs.

Weeks 1, 3, and 4